P30613 · KPYR_HUMAN

Protein
Pyruvate kinase PKLR

Gene
PKLR

Status
UniProtKB reviewed (Swiss-Prot)

Organism
Homo sapiens (Human)

Amino acids
574 (go to sequence)

Protein existence
Evidence at protein level

Annotation score
5/5

Function

Pyruvate kinase that catalyzes the conversion of phosphoenolpyruvate to pyruvate with the synthesis of ATP, and which plays a key role in glycolysis (PubMed:11960989).
Also produces the side product 2-phospholactate which can inhibit fructose-2,6-bisphosphate production (PubMed:27294321).
2-phospholactate can be dephosphorylated by PGP which prevents the inhibition of fructose-2,6-bisphosphate production and allows glycolysis to occur (PubMed:27294321). 2 Publications

Miscellaneous

There are 4 isozymes of pyruvate kinase in mammals: L, R, M1 and M2. L type is major isozyme in the liver, R is found in red cells, M1 is the main form in muscle, heart and brain, and M2 is found in early fetal tissues.

Catalytic activity

Rhea:18157

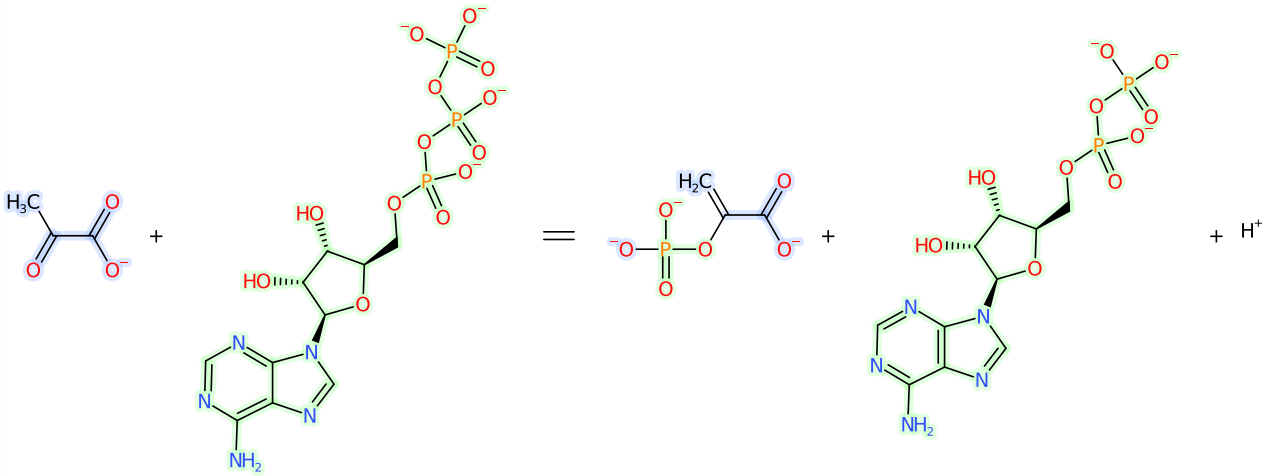
pyruvate + ATP = phosphoenolpyruvate + ADP + H^+ 1 Publication
This reaction proceeds in the backward direction. 1 Publication
EC:2.7.1.40 (UniProtKB | ENZYME | Rhea)

Hide reaction details

Reaction | Atom map

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our Privacy Notice. I agree, dismiss this banner

Color atoms ☑, Number atoms ☐, Confidence = 0.34795



pyruvate » ATP » phosphoenolpyruvate » ADP » H^+ »

Rhea:86567

(S)-lactate + ATP = (2S)-2-phospholactate + ADP + H^+ 1 Publication

View reaction details

Cofactor[i]

Protein has several cofactor binding sites:

Mg^{2+} (UniProtKB | Rhea | CHEBI:18420) 1 Publication
Mn^{2+} (UniProtKB | Rhea | CHEBI:29035) 1 Publication

K^+ (UniProtKB | Rhea | CHEBI:29103) 1 Publication

Activity regulation[i]

Allosterically activated by fructose 1,6-bisphosphate. 1 Publication

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

Features

Showing features for binding site[i], site[i].

±	TYPE	ID	POSITION(S)	DESCRIPTION	
+	Binding site		116	substrate 1 Publication Combined Sources	
+	Binding site		118	K^+ (UniProtKB | ChEBI) 1 Publication Combined Sources	
+	Binding site		118-121	ATP (UniProtKB | ChEBI) By Similarity	Tools Add
+	Binding site		120	K^+ (UniProtKB | ChEBI) 1 Publication Combined Sources	
+	Binding site		156	K^+ (UniProtKB | ChEBI) 1 Publication Combined Sources	
+				K^+ (UniProtKB | ChEBI)	

Expand table

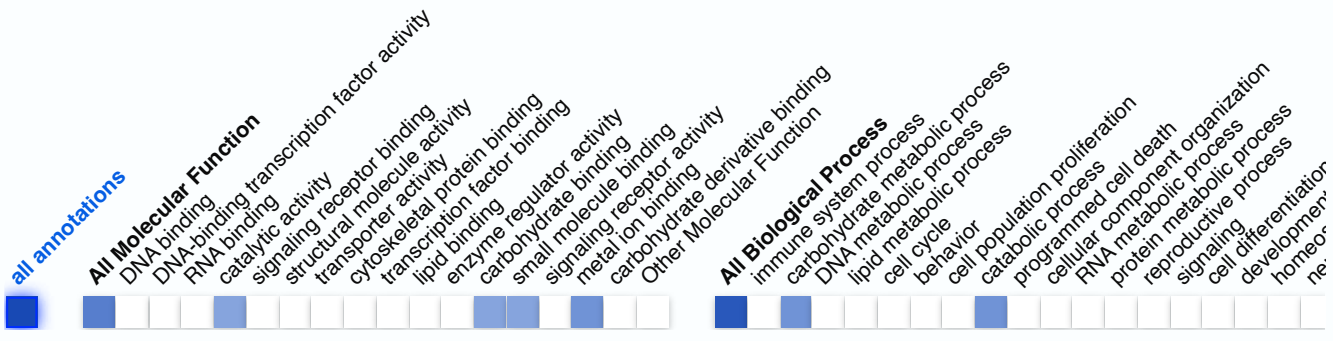
Gene Ontology[i]

GO annotations | GO-CAM models New

Gene Ontology (GO) annotations organized by slimming set.

Slimming set:

Animals and fungi (Opisthokonta)



Cell color indicative of number of GO terms

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

ASPECT	TERM
Molecular Function	monosaccharide binding Source:Ensembl
Molecular Function	potassium ion binding Source:InterPro
Molecular Function	pyruvate kinase activity Source:Reactome 2 Publications
Biological Process	canonical glycolysis Source:Reactome
Biological Process	cellular response to epinephrine stimulus Source:Ensembl
Biological Process	cellular response to insulin stimulus Source:GO_Central
Biological Process	glycolytic process Source:GO_Central
Biological Process	pyruvate biosynthetic process Source:Reactome
Biological Process	response to ATP Source:Ensembl

Expand table

Access the complete set of GO annotations on QuickGO

PAN-GO

P30613 4 GO annotations based on evolutionary models

Keywords[i]

Molecular function

#Allosteric enzyme
#Kinase
#Transferase

Biological process

#Glycolysis

Ligand

#ATP-binding
#Magnesium
#Manganese
#Metal-binding
#Nucleotide-binding

More keywords

Enzyme and pathway databases

BRENDA

2.7.1.40 2681

NDEx

IQUERY-CP-PKLR 7 NDEx IQuery Curated Pathways

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

Reactome
R-HSA-163765 ChREBP activates metabolic gene expression [P30613-1]
R-HSA-210745 Regulation of gene expression in beta cells [P30613-2]
R-HSA-70171 Glycolysis
R-HSA-70268 Pyruvate metabolism
R-HSA-9692914 SARS-CoV-1-host interactions [P30613-2]

SABIO-RK
P30613

SIGNOR
P30613

SignaLink
P30613

UniPathway
UPA00109UER00188

ENZYME
Search...

Names & Taxonomy

Protein names

Recommended name
Pyruvate kinase PKLR

EC number
EC:2.7.1.40 (UniProtKB | ENZYME | Rhea) 1 Publication

Alternative names
Pyruvate kinase 1
Pyruvate kinase isozymes L/R
R-type/L-type pyruvate kinase
Red cell/liver pyruvate kinase

Gene names

Name
PKLR

Synonyms
PK1, PKL

Organism names

Taxonomic identifier
9606 (NCBI)

Organism
Homo sapiens (Human)

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

Accessions[i]

Primary accession
P30613

Secondary accessions
O75758
P11973

Proteomes[i]

Identifier
UP000005640

Component[i]
Chromosome 1

Organism-specific databases

AGR
HGNC:9020

HGNC
HGNC:9020 PKLR

MIM
102900 phenotype
266200 phenotype
609712 gene

VEuPathDB
HostDB:ENSG00000143627

Subcellular Location[i]

UniProt Annotation | GO Annotation

cytosol Source:Reactome
extracellular exosome Source:UniProtKB 1 Publication
pyruvate kinase complex Source:ComplexPortal

Complete GO annotation on QuickGO

Miscellaneous

CD-CODE
91857CE7 Nucleolus

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

Pyruvate kinase hyperactivity (PKHYP)

1 Publication

Note

The disease is caused by variants affecting the gene represented in this entry

Description

Autosomal dominant phenotype characterized by increase of red blood cell ATP.

See also

MIM:102900

Natural variants in PKHYP

VARIANT ID	POSITION(S)	CHANGE	DESCRIPTION
VAR_011435	37	G>E	in PKHYP; dbSNP:rs118204087 1 Publication

Anemia, congenital, non-spherocytic hemolytic, 2 (CNSHA2)

22 Publications

Note

The disease is caused by variants affecting the gene represented in this entry

Description

An autosomal recessive disorder characterized by a variable degree of chronic hemolysis and decreased red cell pyruvate kinase activity. Clinical manifestations range from fatal anemia at birth to a fully compensated hemolysis without apparent anemia.

See also

MIM:266200

Natural variants in CNSHA2

VARIANT ID	POSITION(S)	CHANGE	DESCRIPTION
VAR_058467	40	R>W	in CNSHA2; dbSNP:rs1484388413 1 Publication
VAR_058468	48-53	*missing*	in CNSHA2 1 Publication
VAR_058469	73	L>P	in CNSHA2 1 Publication
VAR_011436	80	S>P	in CNSHA2 1 Publication
VAR_011437	86	R>P	in CNSHA2 1 Publication
VAR_011438	90	I>N	in CNSHA2 1 Publication
VAR_011439	95	G>R	in CNSHA2; dbSNP:rs750857114 1 Publication
VAR_004028	107	M>T	in CNSHA2 1 Publication
VAR_011440	111	G>R	in CNSHA2; dbSNP:rs918627824 1 Publication

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

Expand table

Features

Showing features for natural variant[i].

± ▾	TYPE	ID	POSITION(S)	DESCRIPTION	
+	Natural variant	VAR_011435	37	in PKHYP; dbSNP:rs118204087 1 Publication	
+	Natural variant	VAR_058467	40	in CNSHA2; dbSNP:rs1484388413 1 Publication	
+	Natural variant	VAR_058468	48-53	in CNSHA2 1 Publication	Tools Add
+	Natural variant	VAR_058469	73	in CNSHA2 1 Publication	
+	Natural variant	VAR_011436	80	in CNSHA2 1 Publication	
+	Natural variant	VAR_011437	86	in CNSHA2 1 Publication	
	Natural				

Expand table

Keywords[i]

Disease

#Disease variant

#Hereditary hemolytic anemia

Organism-specific databases

DisGeNET

5313

MIM

102900 phenotype

266200 phenotype

609712 gene

MalaCards

PKLR

OpenTargets

ENSG00000143627

Orphanet

766 Hemolytic anemia due to red cell pyruvate kinase deficiency

Miscellaneous

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

Chemistry

ChEMBL
CHEMBL1075126

DrugBank
DB02726 2-Phosphoglycolic Acid
DB00787 Acyclovir
DB04551 beta-D-fructofuranose 1,6-bisphosphate
DB16236 Mitapivat
DB12835 Olipudase alfa
DB00119 Pyruvic acid

DrugCentral
P30613

GuidetoPHARMACOLOGY
3007

Genetic variation databases

BioMuta
PKLR

ClinGen
HGNC:9020

GenCC
HGNC:9020

Expression[i]

Gene expression databases

Bgee
ENSG00000143627 Expressed in liver and 53 other cell types or tissues

ExpressionAtlas
P30613 baseline and differential

Organism-specific databases

HPA
ENSG00000143627 Tissue enhanced (bone marrow, kidney, liver)

Interaction[i]

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

Binary interactions[i]



TYPE	ENTRY 1	ENTRY 2	NUMBER OF EXPERIMENTS	INTACT
BINARY	P30613	CPNE7 Q9UBL6-2	3	EBI-2117450, EBI-12012272

Complex viewer[i]

Select complex

CPX-3094 PKL pyruvate kinase complex



View interactors in UniProtKB

View CPX-3094 in Complex Portal

Protein-protein interaction databases

BioGRID
111330 65 interactors

CORUM
P30613

ComplexPortal
CPX-3094 PKL pyruvate kinase complex [P30613-2]
CPX-3095 PKR pyruvate kinase complex [P30613-1]

FunCoup
P30613 592 interactors

IntAct
P30613 28 interactors

STRING
9606.ENSP00000339933

Chemistry

BindingDB
P30613

Miscellaneous

RNAct

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

Structure[i]



3D structure databases

AlphaFoldDB
P30613

ModBase
Search...

SMR
P30613

PDBe-KB
Search...

Miscellaneous

EvolutionaryTrace
P30613

Family & Domains[i]

Sequence similarities[i]

Belongs to the **pyruvate kinase family**. Curated

Family and domain databases

- View the Phylogenomic databases for this entry within the **Homologs section**.
- View all family and domain features for this entry's canonical sequence in the **UniParc Feature Viewer**.

CDD
cd00288 Pyruvate_Kinase 1 hit

FunFam
3.20.20.60:FF:000025 Pyruvate kinase 1 hit
3.40.1380.20:FF:000001 Pyruvate kinase 1 hit
2.40.33.10:FF:000023 Pyruvate kinase PKM 1 hit

Gene3D
3.20.20.60 Phosphoenolpyruvate-binding domains 1 hit
2.40.33.10 PK beta-barrel domain-like 1 hit

InterPro
View protein in InterPro
IPR001697 Pyr_Knase
IPR015813 Pyrv/PenolPyrv_kinase-like_dom
IPR040442 Pyrv_kinase-like_dom_sf
IPR011037 Pyrv_Knase-like_insert_dom_sf
More InterPro links

NCBIfam
NF004491 PRK05826.1 1 hit
NF004978 PRK06354.1 1 hit

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

PRINTS
PR01050 PYRUVTKNASE

PROSITE
View protein in PROSITE
PS00110 PYRUVATE_KINASE 1 hit

Pfam
View protein in Pfam
PF00224 PK 1 hit
PF02887 PK_C 1 hit

SUPFAM
SSF51621 Phosphoenolpyruvate/pyruvate domain 1 hit
SSF50800 PK beta-barrel domain-like 1 hit
SSF52935 PK C-terminal domain-like 1 hit

MobiDB
Search…

Sequence & Isoform[i]

Align isoforms (2) **Add isoforms**

Sequence status[i]
Complete

This entry describes **2** isoforms[i] produced by **Alternative splicing**.

P30613-1

This isoform has been chosen as the **canonical** sequence. All positional information in this entry refers to it. This is also the sequence that appears in the downloadable versions of the entry.

Name
R-type
Synonyms
PKR

See also
sequence in **UniParc** or sequence clusters in **UniRef**

Length
574

Mass (Da)
61,830

Last updated
2000-05-30 v2

MD5 Checksum[i]
9E485BFDF0D795CAEA2EE212ABF8C92A

MSIQENISSLQLRSWVSKSQRDLAKSILIGAPGGPAGYLRRASVAQLTQELGTAFFQQQQLPAAMADTFLEHL
CLLDIDSEPVAARSTSIIATIGPASRSVERLKEMIKAGMNIARLNFSHGSHEYHAESIANVREAVESFAGSPLSYR

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

QMLESMITKPRPTRAETSDVANAVLDGADCIMLSGETAKGNFPVEAVKMQHAIAREAEAAVYHRQLFEELRRA
APLSRDPTEVTAIGAVEAAFKCCAAAIIVLTTTGRSAQLLSRYRPRAAVIAVTRSAQAARQVHLCRGVFPLLYRE
PPEAIWADDVDRRVQFGIESGKLRGFLRVGDLVIVVTGWRPGSGYTNIMRVLSIS

P30613-2

Name
L-type
Synonyms
PKL

See also
sequence in **UniParc** or sequence clusters in **UniRef**
Differences from canonical
1-33: MSIQENISSLQLRSWVSKSQRDLAKSILIGAPG → ME Curated

Computationally mapped potential isoform sequences[i]

There are 2 potential isoforms mapped to this entry

Align Add **View all**

Entry	Entry name	Gene name	Length
F8W6W2	F8W6W2_HUMAN	PKLR	161
A0A0G2JLC7	A0A0G2JLC7_HUMAN	PKLR	46

Sequence caution[i]

The sequence **BAA02515.1** differs from that shown. Reason: Erroneous gene model prediction Curated

Features

Showing features for alternative sequence[i], sequence conflict[i].

±	TYPE: All	ID	POSITION(S)	DESCRIPTION	
+	Alternative sequence	VSP_002883	1-33	in isoform **L-type** Curated	Tools Add
+	Sequence conflict		381	in Ref. 3; BAA02515 Curated	
+	Sequence conflict		423	in Ref. 2; AAA60104 Curated	

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

Coding sequence diversity
#Alternative splicing

Technical term
#3D-structure
#Proteomics identification
#Reference proteome

Sequence databases

CCDS
CCDS1109.1 [P30613-1]
CCDS44240.1 [P30613-2]

PIR
I52269 KIHUPR

RefSeq
NP_000289.1 NM_000298.6 [P30613-1]
NP_870986.1 NM_181871.4 [P30613-2]
XP_047278547.1 XM_047422591.1 [P30613-1]
XP_047278548.1 XM_047422592.1 [P30613-1]
XP_054193047.1 XM_054337072.1 [P30613-1]
XP_054193048.1 XM_054337073.1 [P30613-1]

NUCLEOTIDE SEQUENCE	PROTEIN SEQUENCE	MOLECULE TYPE	STATUS
AB015983 EMBL · GenBank · DDBJ	BAA31706.1 EMBL · GenBank · DDBJ	mRNA	
M15465 EMBL · GenBank · DDBJ	AAA60104.1 EMBL · GenBank · DDBJ	mRNA	
D13243 EMBL · GenBank · DDBJ	BAA02515.1 EMBL · GenBank · DDBJ	Genomic DNA	Sequence problems.
AY316591 EMBL · GenBank · DDBJ	AAP69527.1 EMBL · GenBank · DDBJ	Genomic DNA	
BC025737 EMBL · GenBank · DDBJ	AAH25737.1 EMBL · GenBank · DDBJ	mRNA	
S60712 EMBL · GenBank · DDBJ	AAB26262.1 EMBL · GenBank · DDBJ	mRNA	

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

Ensembl
ENST00000342741.6 ENSP00000339933.4 ENSG00000143627.19 [P30613-1]
ENST00000392414.7 ENSP00000376214.3 ENSG00000143627.19 [P30613-2]
ENST00000571194.5 ENSP00000461487.1 ENSG00000262785.5 [P30613-2]
ENST00000572740.1 ENSP00000459921.1 ENSG00000262785.5 [P30613-1]

GeneID
5313

KEGG
hsa:5313

MANE-Select
ENST00000342741.6 ENSP00000339933.4 NM_000298.6 NP_000289.1

UCSC
uc001fka.5 human [P30613-1]

Similar Proteins[i]

UniRef clusters[i]



Homologs[i]

Orthologs & paralogs[i]



Phylogenomic databases

GeneTree
ENSGT00390000008859

HOGENOM
CLU_015439_0_1_1

InParanoid
P30613

OMA
HQGRYDR

OrthoDB
108365at2759

PAN-GO
P30613 4 GO annotations based on evolutionary models

PhylomeDB
P30613

eggNOG
KOG2323 Eukaryota

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

AGR
HGNC:9020

Disclaimer

Any medical or genetic information present in this entry is provided for research, educational and informational purposes only. It is not in any way intended to be used as a substitute for professional medical advice, diagnosis, treatment or care. Our staff consists of biologists and biochemists that are not trained to give medical advice.

© 2002 – 2026 UniProt consortium

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.